

DataStage Dataset (.ds) Hand-off & How to Export the Data

A practical, source-backed guide — with a Databricks Workflow comparison

Prepared by Kiro | June 2026 | All claims verified against IBM documentation and practitioner sources (see Sources).

1. The Scenario: Passing Data Between Jobs/Tasks

In a multi-step pipeline (for example, 3 to 5 jobs or tasks), each step typically writes its result to a persistent location, which then becomes the input for the next step. This is true both in IBM DataStage and in Databricks Workflows, although the underlying mechanics differ.

In DataStage, the recommended hand-off format between parallel jobs is the native **Dataset (.ds)**. So: *Job 1* → *writes employee.ds* → *Job 2 reads it* → *writes employee_clean.ds* → *Job 3 reads it*, with a Sequence Job controlling the run order.

2. What a .ds Dataset Actually Is

A key point that surprises many people: a **.ds file is only a descriptor (pointer) file**, not the data itself. IBM's documentation states that parallel jobs use data sets to store data in persistent form, and that each data set is referred to by a descriptor file, usually with the .ds suffix.

The real data is stored as multiple binary segment files spread across the engine's processing nodes. Per IBM, a data segment contains the records written by a single job run, so a segment can span many partitions, and a partition can hold files from many segments.

- **Descriptor (.ds):** small header file you reference — holds schema and the location of data.
- **Data segments:** the actual records, partitioned and distributed across nodes/disks.
- **Tied to configuration:** layout depends on the engine configuration file (APT_CONFIG_FILE).

Practical warning: never copy, move, or delete .ds files with OS commands (cp, rm). Use the Data Set Management utility or the orchadmin command — otherwise you orphan the underlying data segments.

3. Can You Download the .ds File and Use It Outside DataStage?

Short answer: No. The .ds is proprietary to DataStage and stored in an internal binary format. Copying the descriptor (or even all the segment files) to another machine produces something no external tool — Excel, pandas, Spark, a text editor — can read. Practitioner sources confirm dataset contents cannot be read outside DataStage and are only readable via the Data Set stage. (Content rephrased for licensing compliance.)

4. How to Actually Get the Data Out

To use the data externally, convert it to a portable format. Options, in order of preference:

#	Method	Result / Notes
1	Sequential File stage (CSV / flat file)	Recommended for a full, clean export. Add it as a target inside the job. Writes in parallel for multi
2	Database target stage	Land data into a DB table, then extract using normal SQL/DB tools.
3	orchadmin (command line)	Ships with DataStage; supports delete, copy, describe, and dump. Good for inspecting/automatic
4	Data Set Management (GUI)	Graphical tool in Designer/Director to view, manage, and copy dataset contents.

Recommended approach: give Job 1 a dual output — keep the .ds for the fast, parallel hand-off to Job 2, AND add a Sequential File (CSV) stage so you can download usable data. Example flow: *Source* → *Transformer* → (*Data Set .ds for next job*) + (*Sequential File .csv for export*).

5. DataStage vs. Databricks — Side by Side

Concept	DataStage	Databricks Workflow
Orchestration unit	Sequence Job	Workflow
Step / task	Job	Task (notebook/script)
Big data between tasks	Dataset (.ds), DB table, or flat file	Delta / Parquet in cloud storage
Data flow inside a task	In-memory links (pipeline parallelism)	In-memory DataFrame in code
Small metadata between tasks	Job parameters / user variables	dbutils.jobs.taskValues
Best hand-off format	Dataset (.ds) — keeps partitioning + schema	Delta — ACID, schema, time travel

6. Key Takeaways

- A .ds file is a **descriptor**; the real data is internal, partitioned binary across nodes.
- You **cannot** download a .ds and open it in external tools — it is a proprietary format.
- To export, **convert**: use a Sequential File (CSV) stage or a database target.
- orchadmin dump and the Data Set Management GUI are good for **inspection**, but dump can truncate.
- Conceptually, .ds between DataStage jobs == Delta/Parquet between Databricks tasks.

Sources (Trusted / Verified)

Primary sources are IBM official documentation; practitioner sources are noted as such.

- [1] IBM Documentation — "Developing DataStage and QualityStage parallel jobs" (data sets store data persistently; referred to by a .ds descriptor file). <https://www.ibm.com/docs/en/iis/11.7?topic=qualitystage-developing-parallel-jobs>
- [2] IBM Documentation — "Structure of data sets" (segments and partitions; a segment can contain files from many partitions). <https://www.ibm.com/docs/en/iis/11.5?topic=mds-structure-data-sets>
- [3] IBM Documentation — "Data set stage" (file stage to read/write a data set; parallel or sequential mode). <https://www.ibm.com/docs/en/iis/11.7.0?topic=files-data-sets>
- [4] IBM Documentation — "Sequential file" (executes in parallel for multiple files, sequentially for a single file). https://datapatform.cloud.ibm.com/docs/content/dstage/dsnav/topics/sequential_file_connector.html
- [5] IBM Support — "orchadmin dump command produces truncated output" (caveat: dump output can be truncated). <https://www.ibm.com/support/pages/datastage-orchadmin-dump-command-dataset-produces-truncated-output-which-when-saved-xml-file-extension-has-incorrect-xml-format>
- [6] Practitioner blog — "Processing InfoSphere DataStage DataSet from Command Line (orchadmin)" (orchadmin supports delete, copy, describe, dump; usable for automation). <https://riteshkumargupta.wordpress.com/2012/04/14/processing-infosphere-datastage-dataset-from-command-line-orchadmin-utility/>
- [7] Practitioner community (Spiceworks) — "Reading DataStage Datasets" (dataset contents are in internal format, only readable within DataStage via the Data Set stage). <https://community.spiceworks.com/t/reading-datastage-datasets/888161>
- [8] Practitioner community (Spiceworks) — "How to view/cat the Data Set file from command line" (Data Set Management GUI and command-line viewing). <https://community.spiceworks.com/t/how-to-view-cat-the-data-set-file-from-command-line-in-data-stage/879233>

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